Quadratic Kernel Mapping and Kernel Trick

$a = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

$a^T b = \left[a_1 \;\; a_2 \mid \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \right]^{12} = (a_1 b_1 + a_2 b_2)^2 = a_1^2 b_1^2 + a_2^2 b_2^2 + 2\, a_1 b_1 a_2 b_2$

$[a^T b]^2 = \left[a_1^2 \;\; a_2^2 \;\; \sqrt{2}\, a_1 a_2 \mid \begin{pmatrix} b_1^2 \\ \begin{pmatrix} b_2^2 \\ \sqrt{2}\, b_1 b_2 \end{pmatrix} \end{pmatrix} \right] = \varphi(a)^T \varphi(b)$

$Where: \; \varphi(x) \equiv \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1^2 \\ \begin{pmatrix} x_2^2 \\ \sqrt{2}\, x_1 x_2 \end{pmatrix} \end{pmatrix}^T$

2D $\Rightarrow$ 3D